

Machine Learning: Exercise 1

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Question 1 (Correlation and Independence).

- I. Let X and Y be two independent random variables. Prove that they are also uncorrelated, that is, prove that $\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y]$, or that $\text{cov}(X; Y) = 0$.

Solution. Since $X; Y$ are independent, we have $f_{X;Y}(x; y) = f_X(x) \cdot f_Y(y)$. We have:

$$\begin{aligned}\mathbb{E}[XY] &= \iint_{\mathbb{R}^2} xy f_{X;Y}(x; y) \, dx \, dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) \, dx \, dy \\ &= \int_{-\infty}^{\infty} x f_X(x) \, dx \int_{-\infty}^{\infty} y f_Y(y) \, dy = \mathbb{E}[X] \mathbb{E}[Y].\end{aligned}$$

- II. Give an example of two uncorrelated random variables that are not independent.

Solution. Let X be uniformly distributed on $[-1, 1]$, and define $Y = X^2$.

First, since X is uniformly distributed on $[-1, 1]$ we have $\mathbb{E}[X] = 0$, $\mathbb{E}[Y] = \mathbb{E}[X^2] = 0$, and $\mathbb{E}[XY] = \mathbb{E}[X \cdot X^2] = \mathbb{E}[X^3] = 0$. Therefore, $\text{cov}(X; Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0$.

Second, even though $X; Y$ are uncorrelated, they are clearly dependent (since we define $Y = X^2$).

- III. Give an example of three random variables such that every pair of variables is independent, but any variable can be completely determined if the two other variables are known.

Solution. Let $X; Y$ be independent random variables such that:

$$P(X = 0) = P(X = 1) = \frac{1}{2} \quad \text{and} \quad P(Y = 0) = P(Y = 1) = \frac{1}{2}.$$

Define a third random variable Z by $Z = (X + Y) \bmod 2$ (which is also equivalent to XOR).

We compute:

$$P(X = x; Z = z) = \sum_{y \in \{0; 1\} \text{ s.t. } (x+y) \bmod 2 = z} P(X = x; Y = y),$$

and since $X; Y$ are independent we get $P(X = x; Y = y) = 1/4$. For each fixed $x; z$, there is exactly one y with $z = (x + y) \mod 2$:

$$P(X = x; Z = z) = \frac{1}{4}.$$

Also, $P(X = x) = 1/2$ and $P(Z = z) = 1/2$ (Z is also uniformly distributed on $\{0; 1\}$). Hence:

$$P(X = x; Z = z) = P(X = x) P(Z = z),$$

which means that $X; Z$ are independent.

A similar explanation is available for $Y; Z$ as well.

Although every pair is independent, there is a deterministic relationship among the entire three:

- Knowing $X; Y$ uniquely determines Z .
- Knowing $X; Z$ allows you to recover Y as $Y = (X + Z) \mod 2$.
- Knowing $Y; Z$ allows you to recover X as $X = (Y + Z) \mod 2$.

IV. Let $\mathbf{x}_1; \dots; \mathbf{x}_n$ be a set of column vectors of dimension $d > n$. Let C be an (empirical) population correlation matrix $C = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$. Note that this is not a covariance matrix unless the means are zero.

Prove that that matrix C is positive semi-definite. That is, for any vector $\mathbf{z} \in \mathbb{R}^d$, it holds that $\mathbf{z}^\top C \mathbf{z} \geq 0$.

Solution. For some arbitrary vector $\mathbf{z} \in \mathbb{R}^d$, consider:

$$\mathbf{z}^\top C \mathbf{z} = \mathbf{z}^\top \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right) \mathbf{z}.$$

Since summation is a linear operator, we can exchange the summation with the quadratic form:

$$\mathbf{z}^\top C \mathbf{z} = \sum_{i=1}^n \mathbf{z}^\top (\mathbf{x}_i \mathbf{x}_i^\top) \mathbf{z}.$$

For each i , note that

$$\mathbf{z}^\top (\mathbf{x}_i \mathbf{x}_i^\top) \mathbf{z} = (\mathbf{z}^\top \mathbf{x}_i) (\mathbf{x}_i^\top \mathbf{z}),$$

and, recognising that $\mathbf{z}^\top \mathbf{x}_i$ is a scalar (and of course, equal to its transpose), we have $(\mathbf{z}^\top \mathbf{x}_i) (\mathbf{x}_i^\top \mathbf{z}) = (\mathbf{x}_i^\top \mathbf{z})^2$.

Therefore, we get:

$$\mathbf{z}^\top C \mathbf{z} = \sum_{i=1}^n (\mathbf{x}_i^\top \mathbf{z})^2,$$

and since the squares are all non-zero, the matrix C is positive semi-definite by definition.

Question 2 (Multivariate Gaussians). Let $X \sim \mathcal{N}(\mu; \Sigma)$ be a standard multivariate Gaussian of dimension d . That is, its mean is the vector of zeros $\mu = \mathbf{0}$ and its covariance matrix is the d -dimensional identity matrix $\Sigma = I_d$.

- I. For $d = 1$, what is the probability that a sample is within one standard deviation from the mean (zero)?

Solution. For $d = 1$, the probability that a sample is within one standard deviation from the mean is given by:

$$P(-1 \leq X \leq 1) = \Phi(1) - \Phi(-1).$$

Since the standard normal distribution is symmetric around zero, we have $\Phi(-1) = 1 - \Phi(1)$, and:

$$P(-1 \leq X \leq 1) = \Phi(1) - (1 - \Phi(1)) = 2\Phi(1) - 1 \approx 2 \cdot 0.8413 - 1 \approx 0.6826.$$

- II. Prove that as $d \rightarrow \infty$, the probability that a sample is within a unit distance from the centre decays to zero. Hint, the d -dimensional unit sphere around zero is contained within a d -dimensional cube around zero with an edge size of 2. Use it to bound the probability that a sample is inside the cube in every dimension.

Solution. Let $X \sim \mathcal{N}(\mathbf{0}; I_d)$ be a d -dimensional standard Gaussian random variable, we want to show that:

$$P(\|X\| \leq 1) \xrightarrow{d \rightarrow \infty} 0.$$

First, the unit sphere in $\mathbb{R}^d$, defined as $\{x \in \mathbb{R}^d : \|x\| \leq 1\}$, is completely contained within the hyper-cube $[-1, 1]^d$, since if $\|x\| \leq 1$, then in particular, each coordinate must satisfy $|x_i| \leq 1$.

Also, because the unit sphere is a subset of the hyper-cube, we have:

$$P(\|X\| \leq 1) \leq P(X \in [-1, 1]^d).$$

For a one-dimensional standard Gaussian variable, we found that $P(|X_i| \leq 1) = 2\Phi(1) - 1 \approx 0.6826$, and since the components of X are independent, we have:

$$P(X \in [-1, 1]^d) = \prod_{i=1}^d P(|X_i| \leq 1) = (0.6826)^d,$$

and since $0.6826 < 1$, it follows that $P(X \in [-1, 1]^d) \xrightarrow{d \rightarrow \infty} 0$.

Question 3 (Linear Transformation of the Covariance Matrix). Let X be a random vector $X = [X_1 \ \dots \ X_d]$, with a population covariance matrix C . Let $\mathbf{w}$ be some vector in $\mathbb{R}^d$ and define $y = \mathbf{w}^\top X$. Show that the variance of y is $\mathbf{V}[y] = \mathbf{w}^\top C \mathbf{w}$.

Solution. Let $\mu = \mathbb{E}[X]$. Then, expectation of y satisfies:

$$\mathbb{E}[y] = \mathbf{w}^\top \mathbb{E}[X] = \mathbf{w}^\top \mu,$$

and the variance:

$$\mathbb{V}[y] = \mathbb{E}[(y - \mathbb{E}[y])^2] = \mathbb{E}[(\mathbf{w}^\top (X - \mu))^2].$$

Notice that:

$$(\mathbf{w}^\top (X - \mu))^2 = \mathbf{w}^\top (X - \mu) (X - \mu)^\top \mathbf{w}.$$

Taking expectations and using the linearity of expectation, we have:

$$\mathbb{V}[y] = \mathbf{w}^\top \mathbb{E}[(X - \mu)(X - \mu)^\top] \mathbf{w}.$$

By definition, the population covariance matrix C is:

$$C = \mathbb{E}[(X - \mu)(X - \mu)^\top].$$

Thus, we finally obtain $\mathbb{V}[y] = \mathbf{w}^\top C \mathbf{w}$.

Question 4 (Eigendecomposition). Let A be a square matrix with eigenvalues $\lambda_1; \dots; \lambda_n$.

I. Prove that A and $A^\top$ have the same set of eigenvalues.

Solution. The eigenvalues of A are the roots of its characteristic polynomial $p_A(\lambda) = \det(A - \lambda I)$. Similarly, the eigenvalues of $A^\top$ are the roots of $p_{A^\top}(\lambda) = \det(A^\top - \lambda I)$. Recall the property of determinants that for any square matrix B , we have $\det(B^\top) = \det(B)$, and we have:

$$\det(A^\top - \lambda I) = \det((A - \lambda I)^\top) = \det(A - \lambda I).$$

Since we have:

$$p_{A^\top}(\lambda) = \det(A^\top - \lambda I) = \det(A - \lambda I) = p_A(\lambda),$$

the characteristic polynomials of A ; $A^\top$ are identical, and since the eigenvalues of a matrix are the roots of the characteristic polynomial we get that A ; $A^\top$ have the exact same set of eigenvalues.

II. Assume that A is diagonalisable. Prove that the determinant of A is given by the product of its eigenvalues: $\det(A) = \prod_{i=1}^n \lambda_i$.

Solution. Since A is diagonalisable, there exists an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$, where the diagonal entries of D are the eigenvalues of A , meaning, $D = \text{diag}(\{\lambda_1; \dots; \lambda_n\})$.

The determinant of a product of matrices is the product of their determinants. Therefore:

$$\det(A) = \det(PDP^{-1}) = \det(P) \det(D) \det(P^{-1}) = \det(D).$$

Since D is a diagonal matrix with diagonal entries, its determinant is the product of its diagonal entries:

$$\det(D) = \prod_{i=1}^n \lambda_i,$$

and hence, if A is diagonalisable, its determinant equals to the product of its eigenvalues.

Question 5 (Gradients). Below are two functions of the form $f(\mathbf{x}) : \mathbb{R}^d \rightarrow \mathbb{R}$ that take an input vector $\mathbf{x} \in \mathbb{R}^d$. State the range where the function is differentiable. Compute their gradients with respect to $\mathbf{x}$. In both cases, $\mathbf{w} \in \mathbb{R}^d$ is a given vector of constants.

I.

$$f(\mathbf{w}; \mathbf{x}) = \begin{cases} 0 & \mathbf{w}^\top \mathbf{x} \geq 1, \\ 1 - \mathbf{w}^\top \mathbf{x} & \text{otherwise.} \end{cases}$$

Solution. We consider two regions:

- (a) Region $\mathbf{w}^\top \mathbf{x} > 1$: In this region, the function is defined as $f(\mathbf{x}) = 0$, a constant function, which is differentiable anywhere with gradient = 0.
- (b) Region $\mathbf{w}^\top \mathbf{x} < 1$: In this region, the function is linear with gradient = $-\mathbf{w}$.
- (c) Boundary $\mathbf{w}^\top \mathbf{x} = 1$: Although f is continuous at this boundary, it is not differentiable. See that:

$$\begin{aligned} \lim_{\mathbf{w}^\top \mathbf{x} \rightarrow 1^-} \nabla f(\mathbf{x}) &= -\mathbf{w}, \\ \lim_{\mathbf{w}^\top \mathbf{x} \rightarrow 1^+} \nabla f(\mathbf{x}) &= 0. \end{aligned}$$

The two gradients coincide $\iff -\mathbf{w} = 0$, i.e., in the trivial case. Therefore:

$$\nabla f(\mathbf{x}) = \begin{cases} -\mathbf{w} & \mathbf{w}^\top \mathbf{x} < 1, \\ \mathbf{0} & \mathbf{w}^\top \mathbf{x} > 1, \\ \text{Not differentiable} & \mathbf{w}^\top \mathbf{x} = 1. \end{cases}$$

II.

$$f(\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}.$$

Show that the gradient w.r.t. $\mathbf{x}$ equals to $\nabla f(\mathbf{x}) = f(\mathbf{x})(1 - f(\mathbf{x}))\mathbf{w}$.

Solution. Let $\mathbf{t} = \mathbf{w}^\top \mathbf{x}$. Then, we have $f(\mathbf{t}) = (1 + \exp(-\mathbf{t}))^{-1}$. See that:

$$\nabla f(\mathbf{t}) = (1 + \exp(-\mathbf{t}))^{-2} \cdot (-\exp(-\mathbf{t})) = \frac{\exp(-\mathbf{t})}{(1 + \exp(-\mathbf{t}))^2} = f(\mathbf{t})(1 - f(\mathbf{t})).$$

So, we have $\nabla f(\mathbf{t}) = f(\mathbf{t})(1 - f(\mathbf{t}))$. Apply the chain rule:

$$\nabla f(\mathbf{x}) = \nabla f(\mathbf{t}_{=\mathbf{x}}) \cdot \nabla_{\mathbf{x}} \mathbf{t} = f(\mathbf{x})(1 - f(\mathbf{x})) \cdot \mathbf{w}.$$